

Human Reading

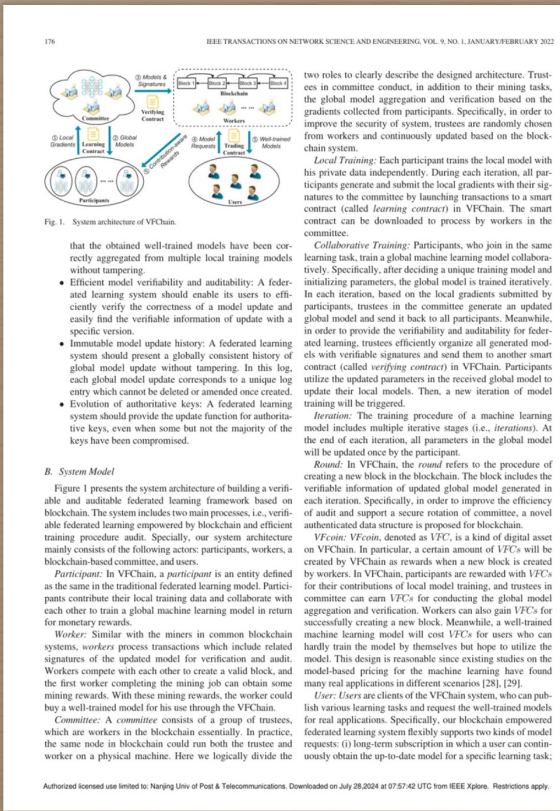


Fig. 1. System architecture of VFChain.

that the obtained well-trained models have been correctly aggregated from multiple local training models without tampering.

- Efficient model verifiability and auditability: A federated learning system should enable its users to efficiently verify the correctness of a model update and easily find the verifiable information of update with a specific version.
- Immutability model update history: A federated learning system should present a globally consistent history of global model update without tampering. In this log, each global model update corresponds to a unique log entry which cannot be deleted or amended once created.
- Evolution of authoritative keys: A federated learning system should provide the update function for authoritative keys, even when some but not the majority of the keys have been compromised.

B. System Model

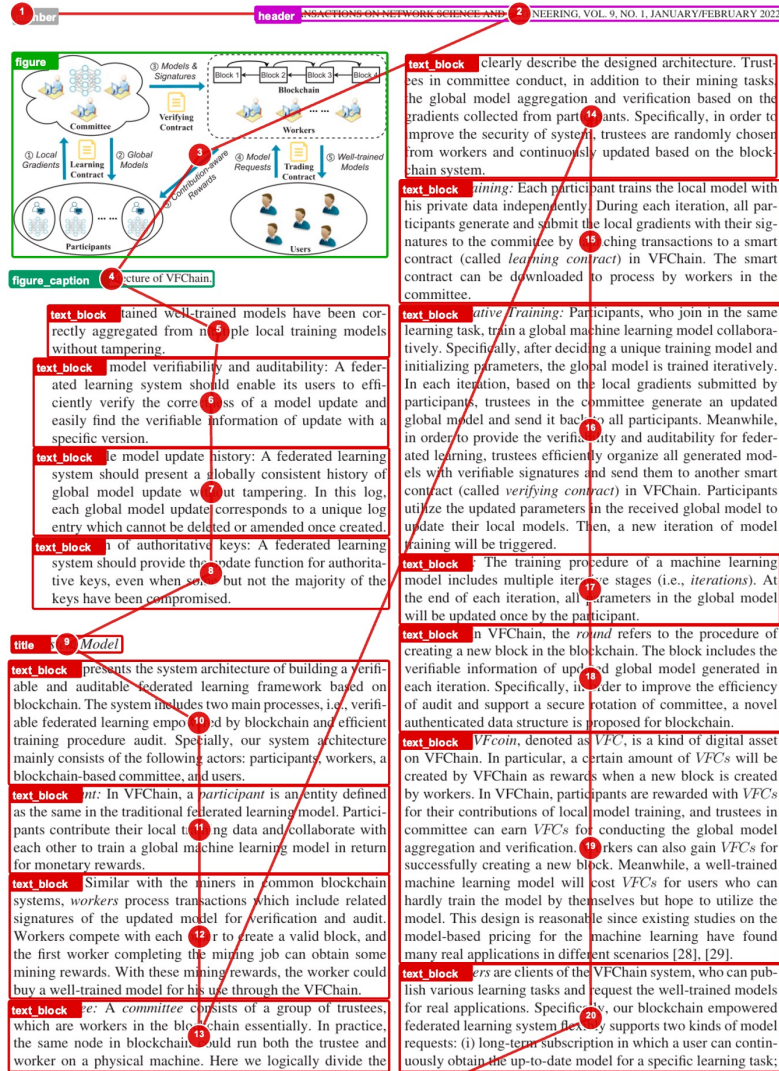
Figure 1 presents the system architecture of building a verifiable and auditable federated learning framework based on blockchain. The system includes two main processes, i.e., verifiable federated learning empowered by blockchain and efficient training procedure audit. Specially, our system architecture mainly consists of the following actors: participants, workers, a blockchain-based committee, and users.

Participant: In VFChain, a *participant* is an entity defined as the same in the traditional federated learning model. Participants contribute their local training data and collaborate with each other to train a global machine learning model in return for monetary rewards.

Worker: Similar with the miners in common blockchain systems, *workers* process transactions which include related signatures of the updated model for verification and audit. Workers compete with each other to create a valid block, and the first worker completing the mining job can obtain some mining rewards. With these mining rewards, the worker could buy a well-trained model for his use through the VFChain.

Committee: A committee consists of a group of trustees, which are workers in the blockchain essentially. In practice, the same node in blockchain could run both the trustee and worker on a physical machine. Here we logically divide the

Ordered-Reading



figure_caption 4 Structure of VFChain.

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Adaptive Attention Allocating

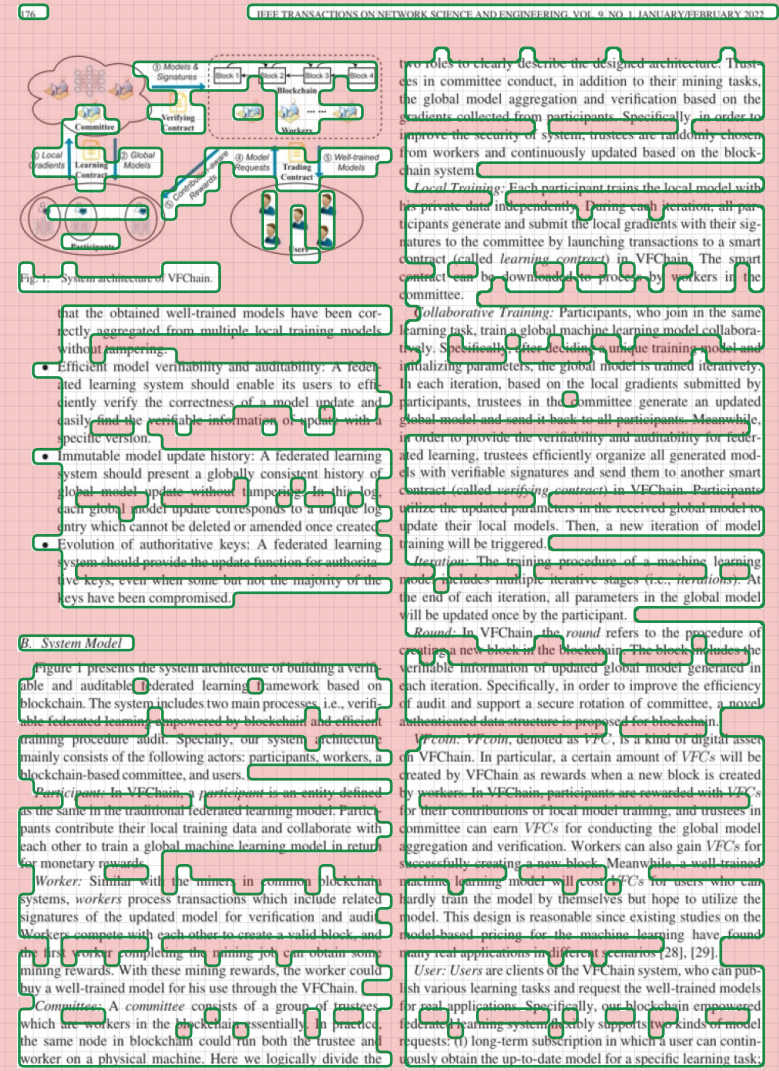


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